

Malkin Textbook Responses – Final Report

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1 2-4 Malkin

Explain why the value θ_K , entering the Kohlrausch function, Eq. 2.1.6 is not a relaxation time. How do you find relaxation times for this relaxation function?

$$\varphi(t) = \exp \left[- \left(\frac{t}{\theta_K} \right)^n \right], \quad \varphi(t) = \int_0^\infty G(\theta) e^{-t/\theta} d\theta.$$

where $G(\theta)$ is the relaxation-time spectrum (what we want to find). Let $s = 1/\theta$, so $d\theta = -ds/s^2$,

$$\varphi(t) = \int_0^\infty \frac{G(1/s)}{s^2} e^{-st} ds \equiv \int_0^\infty \tilde{G}(s) e^{-st} ds, \quad \tilde{G}(s) = \frac{G(1/s)}{s^2}.$$

Therefore $\tilde{G}$ is obtained by an inverse Laplace transform:

$$\tilde{G}(s) = \mathcal{L}^{-1}\{\varphi(t)\}(s) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{st} \varphi(t) dt$$

$$\tilde{G}(s) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{st} e^{-(s_K t)^n} dt, \quad \text{where } s_K = 1/\theta_K$$

But without solving, we can just say:

$$\tilde{G}(s) = \mathcal{L}^{-1}\{\varphi(t)\}(s) \implies G(\theta) = \frac{\tilde{G}(1/\theta)}{\theta^2}.$$

All together

$$G(\theta) = \frac{1}{\theta^2} \mathcal{L}^{-1} \{ e^{-(s_K t)^n} \} (1/\theta).$$

1.1 Discrete relaxation times

If we want to solve it we need to make it discrete. So we can write:

$$G(\theta) = \sum_{i=1}^N G_i \delta(\theta - \theta_i)$$

so that there are T discrete relaxation times θ_i with weights G_i . Then we have our relaxation function as:

$$\varphi(t) = \sum_{i=1}^N \int_0^\infty G_i \delta(\theta - \theta_i) e^{-t/\theta} d\theta = \sum_{i=1}^N G_i e^{-t/\theta_i}.$$

Now we can equate this to the Kohlrausch function:

$$e^{-(t/\theta_K)^n} = \sum_{i=1}^N G_i e^{-t/\theta_i}.$$

1.2 Finishing the inverse Laplace transform

We have

$$\tilde{G}(s) = \mathcal{L}^{-1} \{ e^{-(s_K t)^n} \} (s), \quad s_K = \frac{1}{\theta_K}, \quad 0 < n < 1.$$

Expand the stretched exponential:

$$e^{-(s_K t)^n} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} s_K^{nk} t^{nk}.$$

Use the inverse Laplace transform identity

$$\mathcal{L}^{-1} \{ t^a \} (s) = \frac{1}{\Gamma(-a)} s^{-a-1}$$

which can be obtained by analytic continuation of $\mathcal{L} \{ t^{\alpha-1} / \Gamma(\alpha) \} = s^{-\alpha}$. With $a = nk$, this gives

$$\mathcal{L}^{-1} \{ t^{nk} \} (s) = \frac{1}{\Gamma(-nk)} s^{-nk-1}.$$

Therefore

$$\tilde{G}(s) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} s_K^{nk} \frac{1}{\Gamma(-nk)} s^{-nk-1}.$$

1.3 Convert back to the relaxation-time spectrum

Recall $\tilde{G}(s) = G(1/s)/s^2$, hence $G(\theta) = \tilde{G}(1/\theta)/\theta^2$. Substitute $s = 1/\theta$:

$$\tilde{G}(1/\theta) = \theta \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \frac{1}{\Gamma(-nk)} (s_K \theta)^{nk}.$$

Thus

$$G(\theta) = \frac{1}{\theta} \sum_{k=1}^{\infty} \frac{(-1)^k}{k!} \frac{1}{\Gamma(-nk)} \left(\frac{\theta}{\theta_K} \right)^{nk}, \quad (0 < n < 1). \quad (1.3.1)$$

Does this converge?

$$\int_0^{\infty} G(\theta) d\theta = \sum_{k=1}^{\infty} \frac{(-1)^k}{k! \Gamma(-nk)} \theta_K^{-nk} \int_0^{\infty} \theta^{nk-1} d\theta$$

We know that $0 < n < 1$, so $nk - 1 > -1$ for $k \geq 1$. Therefore the integral diverges:

$$= \sum_{k=1}^{\infty} \frac{(-1)^k}{k! \Gamma(-nk)} \theta_K^{-nk} \cdot \infty$$

1.4 Simulation

When I try to simulate Eq. 1.3.1 it doesn't converge well. This is likely because of the $\Gamma(-nk)$ term in the denominator which diverges for integer nk and therefore isn't numerically stable. To fix this we use the Euler's reflection formula:

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)} \implies \Gamma(-nk) = \frac{-\pi}{\sin(\pi nk)\Gamma(1+nk)}.$$

Substituting this into Eq. 1.3.1 gives:

$$G(\theta) = \frac{1}{\theta} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k!} \frac{\sin(\pi nk)}{\pi} \Gamma(1+nk) \left(\frac{\theta}{\theta_K} \right)^{nk}, \quad (0 < n < 1).$$

We can attempt to show numerically that

$$\int_0^{\infty} G(\theta) e^{-t/\theta} d\theta \approx e^{-(t/\theta_K)^n}$$

After trying this in python it *does* blow up.

2 2-8 Malkin

Eq. 2.3.11 and its solution show that the Burgers model describes the behavior of a material with two relaxation times. The same behavior is represented by two parallel Maxwell elements with their relaxation times $\theta_1 = n_1/G_1$ and $\theta_2 = n_2/G_2$ where n_1 and G_1 are the viscosity and elastic modulus of the first and n_2 and G_2 of the second Maxwell elements joined in parallel. Calculate the values of the constants of the Burgers model expressed via constants of the two Maxwell elements.

2.1 Burgers Model:

For the total strain rate we have:

$$\begin{aligned} \dot{\gamma} &= \dot{\gamma}_m + \dot{\gamma}_k = \frac{\sigma}{\eta_m} + \frac{\dot{\sigma}}{G_m} + \dot{\gamma}_k. \\ \dot{\gamma}_k &= \dot{\gamma} - \frac{\sigma}{\eta_m} - \frac{\dot{\sigma}}{G_m}, \quad \ddot{\gamma}_k = \ddot{\gamma} - \frac{\dot{\sigma}}{\eta_m} - \frac{\ddot{\sigma}}{G_m}. \end{aligned}$$

The stress across the Maxwell and Kelvin-Voigt elements is the same:

$$\sigma = \sigma_m = \sigma_k \implies \sigma = G_k \gamma_k + \eta_k \dot{\gamma}_k.$$

Now we can differentiate this then substitute Eq. 2.1 into it:

$$\dot{\sigma} = G_k \dot{\gamma}_k + \eta_k \ddot{\gamma}_k.$$

$$\dot{\sigma} = G_k \left(\dot{\gamma} - \frac{\sigma}{\eta_m} - \frac{\dot{\sigma}}{G_m} \right) + \eta_k \left(\ddot{\gamma} - \frac{\dot{\sigma}}{\eta_m} - \frac{\ddot{\sigma}}{G_m} \right).$$

Collecting terms gives:

$$\frac{\eta_k}{G_m} \ddot{\sigma} + \left(1 + \frac{G_k}{G_m} + \frac{\eta_k}{\eta_m} \right) \dot{\sigma} + \frac{G_k}{\eta_m} \sigma = \eta_k \ddot{\gamma} + G_k \dot{\gamma}. \quad (2.1.1)$$

2.2 Two Maxwell Elements in Parallel:

Consider branch 1, a Maxwell element with spring G_1 in series with dashpot η_1 . Within this Maxwell element the stress is the same in the spring and dashpot:

$$\sigma_1 = \sigma_{G_1} = \sigma_{\eta_1}, \quad \sigma_1 = G_1 \gamma_{G_1}, \quad \sigma_1 = \eta_1 \dot{\gamma}_{\eta_1}.$$

Therefore,

$$\dot{\gamma}_{G_1} = \frac{\dot{\sigma}_1}{G_1}, \quad \dot{\gamma}_{\eta_1} = \frac{\sigma_1}{\eta_1}.$$

Since the spring and dashpot are in series, strains add in branch 1:

$$\begin{aligned} \dot{\gamma}_1 &= \dot{\gamma}_{G_1} + \dot{\gamma}_{\eta_1} \\ &= \frac{\dot{\sigma}_1}{G_1} + \frac{\sigma_1}{\eta_1}. \end{aligned}$$

Introducing the time-derivative operator $D \equiv \frac{d}{dt}$, this may be written as

$$\dot{\gamma}_1 = \left(\frac{1}{G_1} D + \frac{1}{\eta_1} \right) \sigma_1 = \frac{\eta_1 D + G_1}{G_1 \eta_1} \sigma_1.$$

Similarly for branch 2:

$$\dot{\gamma}_2 = \frac{\dot{\sigma}_2}{G_2} + \frac{\sigma_2}{\eta_2} = \left(\frac{1}{G_2} D + \frac{1}{\eta_2} \right) \sigma_2 = \frac{\eta_2 D + G_2}{G_2 \eta_2} \sigma_2.$$

Since the two branches are in parallel, the total strain rate is the same in both branches:

$$\begin{aligned} \dot{\gamma} &= \frac{\eta_1 D + G_1}{G_1 \eta_1} \sigma_1 = \frac{\eta_2 D + G_2}{G_2 \eta_2} \sigma_2. \\ \sigma_1 &= \frac{G_1 \eta_1}{\eta_1 D + G_1} \dot{\gamma}, \quad \sigma_2 = \frac{G_2 \eta_2}{\eta_2 D + G_2} \dot{\gamma}. \end{aligned}$$

The total stress is the sum of the stresses in each branch:

$$\begin{aligned} \sigma &= \sigma_1 + \sigma_2 \\ \sigma &= \frac{G_1 \eta_1}{\eta_1 D + G_1} \dot{\gamma} + \frac{G_2 \eta_2}{\eta_2 D + G_2} \dot{\gamma} \\ \sigma &= \left(\frac{G_1 \eta_1}{\eta_1 D + G_1} + \frac{G_2 \eta_2}{\eta_2 D + G_2} \right) \dot{\gamma} \\ \sigma &= \frac{G_1 \eta_1 (\eta_2 D + G_2) + G_2 \eta_2 (\eta_1 D + G_1)}{(\eta_2 D + G_2)(\eta_1 D + G_1)} \dot{\gamma} \\ (\eta_2 D + G_2)(\eta_1 D + G_1) \sigma &= (G_1 \eta_1 (\eta_2 D + G_2) + G_2 \eta_2 (\eta_1 D + G_1)) \dot{\gamma} \\ (\eta_1 \eta_2 D^2 + (\eta_1 G_2 + \eta_2 G_1) D + G_1 G_2) \sigma &= (G_1 \eta_1 \eta_2 D + G_1 \eta_1 G_2 + G_2 \eta_2 \eta_1 D + G_2 \eta_2 G_1) \dot{\gamma} \\ \eta_1 \eta_2 \ddot{\sigma} + (\eta_1 G_2 + \eta_2 G_1) \dot{\sigma} + G_1 G_2 \sigma &= ((G_1 \eta_1 \eta_2 + G_2 \eta_2 \eta_1) D + G_1 \eta_1 G_2 + G_2 \eta_2 G_1) \dot{\gamma} \end{aligned}$$

$$\eta_1\eta_2\ddot{\sigma} + (\eta_1G_2 + \eta_2G_1)\dot{\sigma} + G_1G_2\sigma = (G_1\eta_1\eta_2 + G_2\eta_2\eta_1)\ddot{\gamma} + (G_1\eta_1G_2 + G_2\eta_2G_1)\dot{\gamma}$$

Finally,

$$\frac{\eta_1\eta_2}{G_1G_2}\ddot{\sigma} + \frac{(\eta_1G_2 + \eta_2G_1)}{G_1G_2}\dot{\sigma} + \sigma = \frac{\eta_1(G_1\eta_2 + G_2\eta_2)}{G_1G_2}\ddot{\gamma} + (\eta_1 + \eta_2)\dot{\gamma} \quad (2.2.1)$$

using $\theta_i = \eta_i/G_i$:

$$\theta_1\theta_2\ddot{\sigma} + (\theta_1 + \theta_2)\dot{\sigma} + \sigma = (\theta_1 + \theta_2)\ddot{\gamma} + (\theta_1G_1 + \theta_2G_2)\dot{\gamma} \quad (2.2.2)$$

gives us a cleaner form.

2.3 Equating Coefficients

From Eq. 2.2.1 we have

$$\frac{\eta_1\eta_2}{G_1G_2}\ddot{\sigma} + \frac{(\eta_1G_2 + \eta_2G_1)}{G_1G_2}\dot{\sigma} + \sigma = \frac{\eta_1\eta_2(G_1 + G_2)}{G_1G_2}\ddot{\gamma} + (\eta_1 + \eta_2)\dot{\gamma}$$

From Eq. 2.1.1 we have

$$\frac{\eta_k}{G_m}\ddot{\sigma} + \left(1 + \frac{G_k}{G_m} + \frac{\eta_k}{\eta_m}\right)\dot{\sigma} + \frac{G_k}{\eta_m}\sigma = \eta_k\ddot{\gamma} + G_k\dot{\gamma}.$$

So we can compare the two equations, we'll multiply Eq. 2.1.1 by $\frac{\eta_m}{G_k}$:

$$\frac{\eta_k\eta_m}{G_mG_k}\ddot{\sigma} + \left(\frac{\eta_m}{G_k} + \frac{\eta_m}{G_m} + \frac{\eta_k}{G_k}\right)\dot{\sigma} + \sigma = \frac{\eta_k\eta_m}{G_k}\ddot{\gamma} + \eta_m\dot{\gamma}.$$

From left to right, we equate coefficients:

$$\frac{\eta_1\eta_2}{G_1G_2} = \frac{\eta_k\eta_m}{G_mG_k} \quad (2.3.1)$$

$$\frac{(\eta_1G_2 + \eta_2G_1)}{G_1G_2} = \frac{\eta_m}{G_k} + \frac{\eta_m}{G_m} + \frac{\eta_k}{G_k} \quad (2.3.2)$$

$$\frac{\eta_1\eta_2(G_1 + G_2)}{G_1G_2} = \frac{\eta_k\eta_m}{G_k} \quad (2.3.3)$$

$$(\eta_1 + \eta_2) = \eta_m \quad (2.3.4)$$

Firstly we have the trivial result from Eq. 2.3.4:

$$\boxed{\eta_m = \eta_1 + \eta_2}.$$

Theoretically, we also have

$$G_m = G_1 + G_2$$

but we can show this by plugging Eq. 2.3.1 into Eq. 2.3.3:

$$\begin{aligned} \frac{\eta_1\eta_2}{G_1G_2}(G_1 + G_2) &= \frac{\eta_k\eta_m}{G_k} \\ \frac{\eta_k\eta_m}{G_mG_k}(G_1 + G_2) &= \frac{\eta_k\eta_m}{G_k} \end{aligned}$$

$$\boxed{G_m = G_1 + G_2}.$$

Using $\eta_m = \eta_1 + \eta_2$, $G_m = G_1 + G_2$, and

$$\frac{\eta_k}{G_k} = \frac{\eta_1 \eta_2 (G_1 + G_2)}{G_1 G_2 (\eta_1 + \eta_2)},$$

we can substitute into Eq. 2.3.2 to solve for G_k :

$$\begin{aligned} \frac{\eta_m}{G_k} &= \frac{\eta_1 G_2 + \eta_2 G_1}{G_1 G_2} - \frac{\eta_1 + \eta_2}{G_1 + G_2} - \frac{\eta_1 \eta_2 (G_1 + G_2)}{G_1 G_2 (\eta_1 + \eta_2)}. \\ \frac{\eta_m}{G_k} &= \frac{(\eta_1 G_2 + \eta_2 G_1)(G_1 + G_2)(\eta_1 + \eta_2) - G_1 G_2 (\eta_1 + \eta_2)^2 - \eta_1 \eta_2 (G_1 + G_2)^2}{G_1 G_2 (G_1 + G_2)(\eta_1 + \eta_2)}. \end{aligned}$$

Now expand the first term in the numerator:

$$\begin{aligned} (\eta_1 G_2 + \eta_2 G_1)(\eta_1 + \eta_2) &= \eta_1^2 G_2 + \eta_1 \eta_2 G_2 + \eta_1 \eta_2 G_1 + \eta_2^2 G_1 \\ &= \eta_1^2 G_2 + \eta_2^2 G_1 + \eta_1 \eta_2 (G_1 + G_2). \end{aligned}$$

So

$$\begin{aligned} (\eta_1 G_2 + \eta_2 G_1)(G_1 + G_2)(\eta_1 + \eta_2) &= (G_1 + G_2) [\eta_1^2 G_2 + \eta_2^2 G_1 + \eta_1 \eta_2 (G_1 + G_2)] \\ &= (G_1 + G_2) (\eta_1^2 G_2 + \eta_2^2 G_1) + \eta_1 \eta_2 (G_1 + G_2)^2. \end{aligned}$$

Substitute this back into the numerator; the $\eta_1 \eta_2 (G_1 + G_2)^2$ terms cancel:

$$\begin{aligned} \text{numerator} &= \left[(G_1 + G_2) (\eta_1^2 G_2 + \eta_2^2 G_1) + \eta_1 \eta_2 (G_1 + G_2)^2 \right] \\ &\quad - G_1 G_2 (\eta_1 + \eta_2)^2 - \eta_1 \eta_2 (G_1 + G_2)^2 \\ &= (G_1 + G_2) (\eta_1^2 G_2 + \eta_2^2 G_1) - G_1 G_2 (\eta_1 + \eta_2)^2. \end{aligned}$$

Expand the remaining pieces:

$$\begin{aligned} (G_1 + G_2) (\eta_1^2 G_2 + \eta_2^2 G_1) &= \eta_1^2 G_1 G_2 + \eta_1^2 G_2^2 + \eta_2^2 G_1^2 + \eta_2^2 G_1 G_2, \\ G_1 G_2 (\eta_1 + \eta_2)^2 &= G_1 G_2 (\eta_1^2 + 2\eta_1 \eta_2 + \eta_2^2) = \eta_1^2 G_1 G_2 + 2\eta_1 \eta_2 G_1 G_2 + \eta_2^2 G_1 G_2. \end{aligned}$$

Subtracting gives

$$\text{numerator} = (\eta_1^2 G_2^2 + \eta_2^2 G_1^2) - 2\eta_1 \eta_2 G_1 G_2 = (G_1 \eta_2 - G_2 \eta_1)^2.$$

then

$$\frac{\eta_m}{G_k} = \frac{(G_1 \eta_2 - G_2 \eta_1)^2}{G_1 G_2 (G_1 + G_2) (\eta_1 + \eta_2)}.$$

Since $\eta_m = \eta_1 + \eta_2$, invert to obtain

$$\boxed{G_k = \frac{G_1 G_2 (G_1 + G_2) (\eta_1 + \eta_2)^2}{(G_1 \eta_2 - G_2 \eta_1)^2}}.$$

Finally, using Eq. 2.3.1 $\eta_k/G_k = \eta_1 \eta_2 (G_1 + G_2)/[G_1 G_2 (\eta_1 + \eta_2)]$, we can solve for η_k :

$$\eta_k = G_k \frac{\eta_k}{G_k} = \left[\frac{G_1 G_2 (G_1 + G_2) (\eta_1 + \eta_2)^2}{(G_1 \eta_2 - G_2 \eta_1)^2} \right] \left[\frac{\eta_1 \eta_2 (G_1 + G_2)}{G_1 G_2 (\eta_1 + \eta_2)} \right],$$

so

$$\boxed{\eta_k = \frac{\eta_1 \eta_2 (G_1 + G_2)^2 (\eta_1 + \eta_2)}{(G_1 \eta_2 - G_2 \eta_1)^2}}.$$

3 3-5 and 3-6 Malkin

3-5: Calculate shear stresses in the flow of liquid through a straight tube if the flow is created by the pressure gradient $\Delta p/L$ (L is the length of a tube).

3-6: Calculate the radial distribution of shear rates and flow velocity of Newtonian liquid (having viscosity η) through a straight tube with radius R .

Starting with the Navier-Stokes equation:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \nabla \cdot \boldsymbol{\sigma}.$$

Evaluating the left-hand side, we have the following due to the steady flow assumption:

$$\frac{\partial \mathbf{u}}{\partial t} = 0.$$

And for the other LHS term:

$$((\mathbf{u} \cdot \nabla) \mathbf{u})_i = u_j \partial_j u_i = u_r \partial_r u_i + u_\theta \frac{1}{r} \partial_\theta u_i + u_z \partial_z u_i$$

$u_r = u_\theta = 0$ therefore

$$((\mathbf{u} \cdot \nabla) \mathbf{u})_r = ((\mathbf{u} \cdot \nabla) \mathbf{u})_\theta = 0$$

$\mathbf{u} = u_z(r)$

$$((\mathbf{u} \cdot \nabla) \mathbf{u})_i = u_r \partial_r u_z + u_\theta \frac{1}{r} \partial_\theta u_z + u_z \partial_z u_z = 0$$

Therefore the whole LHS of the Navier Stokes is zero. Navier-Stokes equation reduces to

$$\nabla p = \nabla \cdot \boldsymbol{\sigma}.$$

In cylindrical coordinates, the RHS is

$$\nabla \cdot \boldsymbol{\sigma} = \frac{1}{r} \frac{\partial}{\partial r} (r \sigma_{rz}) + \frac{1}{r} \frac{\partial \sigma_{\theta z}}{\partial \theta} + \frac{\partial \sigma_{zz}}{\partial z} = \frac{1}{r} \frac{\partial}{\partial r} (r \sigma_{rz}).$$

All other terms are zero since $\sigma_{\theta z} = 0$ and σ_{zz} is independent of z for steady flow. The pressure gradient is constant, so $\nabla p = \frac{\Delta P}{L}$, and we have

$$\begin{aligned} \frac{\Delta P}{L} &= \frac{1}{r} \frac{\partial}{\partial r} (r \sigma_{rz}) \\ \frac{\Delta P}{L} r &= \frac{\partial}{\partial r} (r \sigma_{rz}) \\ \int \frac{\Delta P}{L} r dr &= \int \frac{\partial}{\partial r} (r \sigma_{rz}) dr \\ \frac{\Delta P}{L} \frac{r^2}{2} + C_1 &= r \sigma_{rz}. \end{aligned}$$

Solving for the shear stress gives

$$\sigma_{rz} = \frac{\Delta P}{2L} r + \frac{C_1}{r}.$$

The stress must remain finite at the centerline $r = 0$, which requires $C_1 = 0$. Therefore

$$\sigma_{rz} = \frac{\Delta P}{2L} r.$$

This is a key result: the shear stress σ_{rz} is proportional to the radius r and does not depend on the viscosity η .

Now we can use the constitutive relation for a Newtonian fluid $\sigma_{rz} = \eta \dot{\gamma}_{rz}$ to find the shear rate distribution:

$$\dot{\gamma}_{rz} = \frac{\sigma_{rz}}{\eta} = \frac{\Delta P}{2\eta L} r.$$

Integrating the shear rate gives the velocity distribution:

$$\begin{aligned} u_z(r) &= \int \dot{\gamma}_{rz} dr \\ &= \int \frac{\Delta P}{2\eta L} r dr \\ &= \frac{\Delta P}{4\eta L} r^2 + C_2. \end{aligned}$$

Applying the no-slip boundary condition $u_z(R) = 0$ gives $C_2 = -\frac{\Delta P}{4\eta L} R^2$, so the velocity profile becomes

$$u_z(r) = \frac{\Delta P}{4\eta L} (r^2 - R^2).$$

Additional question 1. Calculate the volume output, Q , for the flow of Newtonian liquid.

Additional question 2. Express maximum shear rate via volume output.

Additional question 3. Is the last expression valid for a liquid with arbitrary rheological properties?

4 3-9 Malkin

Analyze the flow of a viscoplastic (“Bingham-type”) liquid through a straight tube of radius R . Find radial stress and velocity distributions and calculate volume output as a function of the pressure gradient.

According to Malkin, the *Bingham Equation* is

$$\sigma = \sigma_Y + \mu \dot{\gamma}, \tag{4.0.1}$$

where σ_Y is the yield stress and μ is the plastic viscosity. Or equivalently,

$$\dot{\gamma} = \begin{cases} 0 & \text{if } \sigma < \sigma_Y \\ \frac{\sigma - \sigma_Y}{\mu} & \text{if } \sigma \geq \sigma_Y \end{cases}.$$

Next we want to find the shear stress distribution σ_{ij} as a function of r . Starting with the *Navier-Stokes equation*:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \nabla \cdot \boldsymbol{\sigma}.$$

Evaluating the left-hand side, we have the following due to the steady flow assumption:

$$\frac{\partial \mathbf{u}}{\partial t} = 0.$$

And for the other LHS term:

$$((\mathbf{u} \cdot \nabla) \mathbf{u})_i = u_j \partial_j u_i = u_r \partial_r u_i + u_\theta \frac{1}{r} \partial_\theta u_i + u_z \partial_z u_i$$

$u_r = u_\theta = 0$ therefore

$$((\mathbf{u} \cdot \nabla) \mathbf{u})_r = ((\mathbf{u} \cdot \nabla) \mathbf{u})_\theta = 0$$

$\mathbf{u} = u_z(r)$

$$((\mathbf{u} \cdot \nabla) \mathbf{u})_i = u_r \partial_r u_z + u_\theta \frac{1}{r} \partial_\theta u_z + u_z \partial_z u_z = 0$$

Therefore the whole LHS of the Navier Stokes is zero. Navier-Stokes equation reduces to

$$\nabla p = \nabla \cdot \boldsymbol{\sigma}.$$

In cylindrical coordinates, the RHS is

$$\nabla \cdot \boldsymbol{\sigma} = \frac{1}{r} \frac{\partial}{\partial r} (r \sigma_{rz}) + \frac{1}{r} \frac{\partial \sigma_{\theta z}}{\partial \theta} + \frac{\partial \sigma_{zz}}{\partial z} = \frac{1}{r} \frac{\partial}{\partial r} (r \sigma_{rz}).$$

All other terms are zero since $\sigma_{\theta z} = 0$ and σ_{zz} is independent of z for steady flow. The pressure gradient is constant, so $\nabla p = \frac{\Delta P}{L}$, and we have

$$\begin{aligned} \frac{\Delta P}{L} &= \frac{1}{r} \frac{\partial}{\partial r} (r \sigma_{rz}) \\ \int \frac{\Delta P}{L} r dr &= \int \frac{\partial}{\partial r} (r \sigma_{rz}) dr \\ \frac{\Delta P}{L} \frac{r^2}{2} + C_1 &= r \sigma_{rz} \\ \sigma_{rz} &= \frac{\Delta P}{2L} r + \frac{C_1}{r}. \end{aligned}$$

The stress must remain finite at the centerline $r = 0$, which requires $C_1 = 0$. Therefore

$$\boxed{\sigma_{rz} = \frac{\Delta P}{2L} r}.$$

Flow only occurs when $\sigma(r) \geq \sigma_Y$, so there is a region in the center of the tube where $r < r_0$ for which there is no shearing. The radius r_0 is given by:

$$\sigma(r_0) = \sigma_Y \implies r_0 = \frac{2L}{\Delta P} \sigma_Y.$$

Recalling that for our “Bingham fluid” $\sigma = \sigma_Y + \mu\dot{\gamma}$, we can write $\dot{\gamma}$ as

$$\dot{\gamma} = \frac{\sigma - \sigma_Y}{\mu} = \frac{\Delta P}{2\mu L} r - \frac{\sigma_Y}{\mu}.$$

Adding the requirement that $\dot{\gamma} = 0$ for $r < r_0$ gives the full shear rate distribution:

$$\dot{\gamma} = \begin{cases} 0 & r < r_0 \\ \frac{\Delta P}{2\mu L} r - \frac{\sigma_Y}{\mu} & r \geq r_0 \end{cases}.$$

As you increase r , the velocity u_z decreases, so the shear rate $\dot{\gamma}$ is negative. Therefore,

$$\dot{\gamma} = -\frac{du_z}{dr} \implies du_z = -\dot{\gamma} dr.$$

Now we can integrate to get the velocity distribution in the region $r \geq r_0$:

$$u_z = -\int \dot{\gamma} dr = -\int \left(\frac{\Delta P}{2\mu L} r - \frac{\sigma_Y}{\mu} \right) dr = -\frac{\Delta P}{4\mu L} r^2 + \frac{\sigma_Y}{\mu} r + C.$$

From the zero slip condition at the wall, we have

$$0 = -\frac{\Delta P}{4\mu L} R^2 + \frac{\sigma_Y}{\mu} R + C \implies C = -\frac{\sigma_Y}{\mu} R + \frac{\Delta P}{4\mu L} R^2.$$

Therefore, the velocity distribution in the region $r \geq r_0$ is

$$u_z(r) = \frac{\Delta P}{4\mu L} (R^2 - r^2) - \frac{\sigma_Y}{\mu} (R - r).$$

For the region $r < r_0$, there is no shearing, so the velocity is constant. The value of this constant is given by the velocity at $r = r_0$:

$$u_z(r) = \frac{\Delta P}{4\mu L} (R^2 - r_0^2) - \frac{\sigma_Y}{\mu} (R - r_0).$$

Recall that

$$r_0 = \frac{2L}{\Delta P} \sigma_Y \implies \sigma_Y = \frac{\Delta P}{2L} r_0,$$

so

$$\frac{\sigma_Y}{\mu}(R - r_0) = \frac{\Delta P}{2\mu L} r_0 (R - r_0).$$

Substituting this into the velocity expression gives

$$\begin{aligned} u_z(r) &= \frac{\Delta P}{4\mu L}(R^2 - r_0^2) - \frac{\Delta P}{2\mu L} r_0 (R - r_0) = \frac{\Delta P}{4\mu L} [(R^2 - r_0^2) - 2r_0(R - r_0)] \\ &= \frac{\Delta P}{4\mu L}(R^2 - 2Rr_0 + r_0^2) = \frac{\Delta P}{4\mu L}(R - r_0)^2. \end{aligned}$$

Then the full velocity distribution is

$$u_z(r) = \begin{cases} \frac{\Delta P}{4\mu L}(R - r_0)^2 & r < r_0 \\ \frac{\Delta P}{4\mu L}(R^2 - r^2) - \frac{\sigma_Y}{\mu}(R - r) & r \geq r_0 \end{cases}.$$

Finally, we can calculate the volume flow Q by integrating the velocity profile across the cross-sectional area of the tube. Since $d\mathbf{A} = r dr d\theta \hat{z}$ and $\mathbf{u} = u_z(r)\hat{z}$, we have

$$\begin{aligned} Q &= 2\pi \int_0^R u_z(r) r dr \\ &= 2\pi \left[\int_0^{r_0} \frac{\Delta P}{4\mu L}(R - r_0)^2 r dr + \int_{r_0}^R \left(\frac{\Delta P}{4\mu L}(R^2 - r^2) - \frac{\sigma_Y}{\mu}(R - r) \right) r dr \right]. \end{aligned}$$

For the region $(0 \leq r \leq r_0)$:

$$\begin{aligned} \int_0^{r_0} \frac{\Delta P}{4\mu L}(R - r_0)^2 r dr &= \frac{\Delta P}{4\mu L}(R - r_0)^2 \int_0^{r_0} r dr \\ &= \frac{\Delta P}{4\mu L}(R - r_0)^2 \left[\frac{r^2}{2} \right]_0^{r_0} \\ &= \frac{\Delta P}{8\mu L}(R - r_0)^2 r_0^2. \end{aligned}$$

For the region $(r_0 \leq r \leq R)$:

$$\int_{r_0}^R \left(\frac{\Delta P}{4\mu L}(R^2 - r^2) - \frac{\sigma_Y}{\mu}(R - r) \right) r dr = \int_{r_0}^R \left[\frac{\Delta P}{4\mu L}(R^2 r - r^3) - \frac{\sigma_Y}{\mu}(Rr - r^2) \right] dr.$$

Integrate term-by-term:

$$\begin{aligned} \int R^2 r dr &= R^2 \frac{r^2}{2}, & \int r^3 dr &= \frac{r^4}{4}, \\ \int Rr dr &= R \frac{r^2}{2}, & \int r^2 dr &= \frac{r^3}{3}. \end{aligned}$$

Plugging the integrals back in gives

$$\begin{aligned} & \int_{r_0}^R \left(\frac{\Delta P}{4\mu L} (R^2 - r^2) - \frac{\sigma_Y}{\mu} (R - r) \right) r \, dr \\ &= \left[\frac{\Delta P}{4\mu L} \left(R^2 \frac{r^2}{2} - \frac{r^4}{4} \right) - \frac{\sigma_Y}{\mu} \left(R \frac{r^2}{2} - \frac{r^3}{3} \right) \right]_{r_0}^R. \end{aligned}$$

Evaluate at $r = R$:

$$\begin{aligned} & \frac{\Delta P}{4\mu L} \left(\frac{R^4}{2} - \frac{R^4}{4} \right) - \frac{\sigma_Y}{\mu} \left(\frac{R^3}{2} - \frac{R^3}{3} \right) \\ &= \frac{\Delta P}{16\mu L} R^4 - \frac{\sigma_Y}{6\mu} R^3. \end{aligned}$$

Evaluate at $r = r_0$:

$$\frac{\Delta P}{4\mu L} \left(R^2 \frac{r_0^2}{2} - \frac{r_0^4}{4} \right) - \frac{\sigma_Y}{\mu} \left(R \frac{r_0^2}{2} - \frac{r_0^3}{3} \right).$$

Combining both regions gives the final expression for Q :

$$Q = 2\pi \left[\frac{\Delta P}{8\mu L} (R - r_0)^2 r_0^2 + \frac{\Delta P}{16\mu L} R^4 - \frac{\sigma_Y}{6\mu} R^3 - \frac{\Delta P}{4\mu L} \left(R^2 \frac{r_0^2}{2} - \frac{r_0^4}{4} \right) + \frac{\sigma_Y}{\mu} \left(R \frac{r_0^2}{2} - \frac{r_0^3}{3} \right) \right]$$

where $r_0 = \frac{2L\sigma_Y}{\Delta P}$, in terms of the pressure gradient $\Delta P/L$, the yield stress σ_Y , the plastic viscosity μ , and the tube radius R .

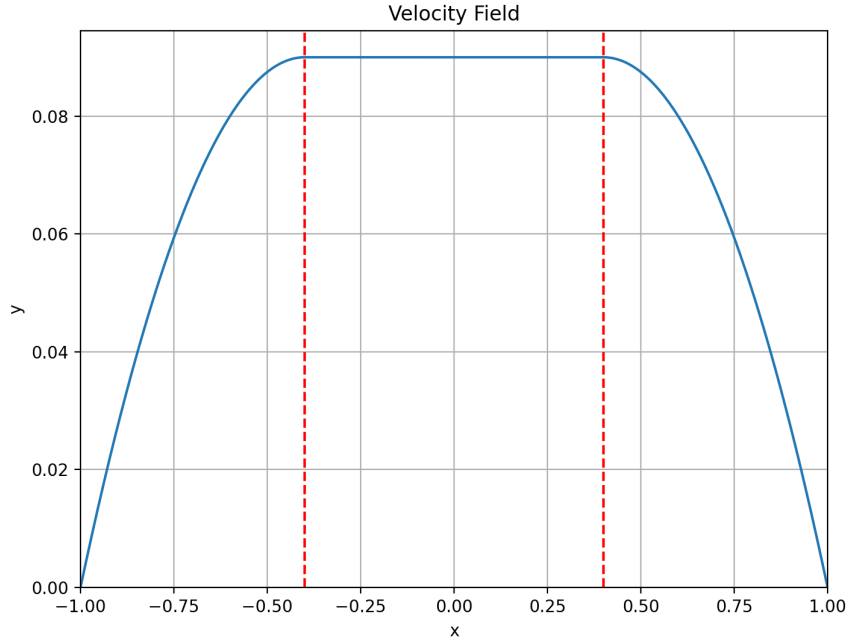


Figure 1: Velocity field for a Bingham fluid flowing through a tube.

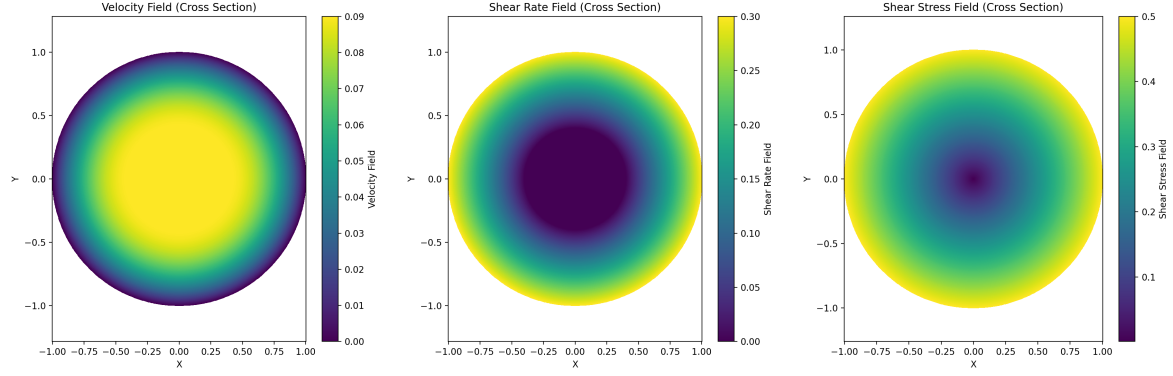


Figure 2: Cross-sectional profiles of velocity, shear rate, and shear stress for a Bingham fluid flowing through a tube. The region near the center of the tube (where $r < r_0$) has zero shear rate. The velocity profile is flat in this region and then decreases towards the walls of the tube. The shear stress increases linearly with radius and exceeds the yield stress σ_Y at $r = r_0$.

5 5-7 Malkin

In the principal scheme of measuring viscoelastic properties of the material, a spring is used (Fig. 5.7.1) that is not ideal elastic but viscoelastic (has some losses in deformation). How do you calculate the viscoelastic properties of the material under investigation?

The spring and sample are in series so the stress σ is the same but the total deformation is the sum:

$$\gamma_{\text{total}} = \gamma_{\text{spring}} + \gamma_{\text{sample}}.$$

For a strain $\gamma = \gamma_0 e^{i\omega t}$, the stress in a linear viscoelastic material is:

$$\sigma = G^* \gamma = (G' + iG'')\gamma_0 e^{i\omega t} \implies \frac{\gamma}{\sigma} = \frac{1}{G^*}.$$

Dividing through by σ gives the compliances:

$$\frac{\gamma_{\text{total}}}{\sigma} = \frac{\gamma_{\text{spring}}}{\sigma} + \frac{\gamma_{\text{sample}}}{\sigma} \implies \frac{1}{G_{\text{total}}^*} = \frac{1}{G_{\text{spring}}^*} + \frac{1}{G_{\text{sample}}^*}.$$

Then we can solve for the compliance of the material in terms of the measured compliance and then compliance of the spring.

$$G_{\text{sample}}^* = \frac{G_{\text{spring}}^* G_{\text{total}}^*}{G_{\text{spring}}^* - G_{\text{total}}^*} \quad (5.0.1)$$

If we replace the spring with a perfectly rigid object, the correction should no longer be needed.

$$G_{\text{total}}^* \lim_{G_{\text{spring}}^* \rightarrow \infty} \frac{G_{\text{total}}^*}{1 - G_{\text{total}}^*/G_{\text{spring}}^*} = G_{\text{total}}^* \quad (5.0.2)$$

and it does.

There's a change all of this is wrong. Malkin does it in a weird way. Maybe this is not exactly what he does but it's something like this. For a rigid system

$$m\ddot{x} + Z\dot{x} + kx = 0$$

Where k and Z are the storage and the loss modules of the system.